Proof. By the relation between permutations and combinations, we have

$$^nC_r = \frac{^nP_r}{r!} = \frac{n!}{r!(n-r)!} \text{ (by a result from permutation).}$$

■

4.5.1 Properties of Combinations

Property 1: (i) $^nC_0 = 1$, (ii) $^nC_n = 1$, (iii) $^nC_r = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}$.

Proof.

$$\begin{aligned}
(i)^nC_0 &= \frac{n!}{0!(n-0)!} = 1.\\
(ii)^nC_n &= \frac{n!}{n!(n-n)!} = \frac{n!}{n!0!} = 1.\\
(iii)^nC_r &= \frac{n(n-1)(n-2)\cdots(n-(r-1))}{r!} = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}.
\end{aligned}$$

■

In view of,

$$^nC_{n-r} = \frac{n!}{(n-r)!(n-(n-r))!} = \frac{n!}{(n-r)!r!} = ^n C_r,$$

we have,

Property 2:

$$^nC_r = ^n C_{n-r}.$$

Property 3: If $^nC_x = ^n C_y$ then either $x = y$ or $x + y = n$.

Proof. By the property 2 we have, $^nC_y = ^n C_{n-y}$

Therefore, $^nC_x = ^n C_y = ^n C_{n-y}$ gives us $x = y$ or $x = n - y$ equivalently, $x = y$ or $x + y = n$. ■

Property 4: $^nC_r + ^n C_{r-1} = ^{n+1} C_r$.

Proof. Using the expressions for the "combination" we have,

$$\begin{aligned}
^nC_r + ^n C_{r-1} &= \frac{n!}{r!.(n-r)!} + \frac{n!}{(r-1)!.(n-(r-1))!}\\
&= \frac{n!}{r! \times (n-r)!} + \frac{n!}{(r-1)! \times (n-r+1)!}\\
&= \frac{n!}{r.(r-1)! \times (n-r)!} + \frac{n!}{(r-1)!.(n-r)!(n-r+1)}\\
&= \frac{n!}{(r-1)! \times (n-r)!} \times \left(\frac{1}{r} + \frac{1}{(n-r+1)}\right)\\
&= \frac{n!}{(r-1)! \times (n-r)!} \times \frac{(n-r+1+r)}{r(n-r+1)}\\
&= \frac{n!}{(r-1)! \times (n-r)!} \times \frac{(n+1)}{r(n-r+1)}\\
&= \frac{(n+1)!}{r! \times (n+1-r)!} = ^{n+1} C_r.
\end{aligned}$$

■